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Citation: [Journal of Mathematical Physics](#) **22**, 2080 (1981); doi: 10.1063/1.525137

View online: <http://dx.doi.org/10.1063/1.525137>

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# Spectral estimates, stability conditions, and the rotating screw-pinch

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(Received 7 August 1980; accepted for publication 17 April 1981)

This article presents two sufficient conditions for the linear stability of rotating ideal plasmas, the first based on conservation of circulation and the second based on circle theorems applicable to linear Hamiltonian systems. The circle theorems also provide bounds on eigenmodes in the complex plane. All results are applied to the rotating screw-pinch which can be described by a single second-order ordinary differential equation.

PACS numbers: 52.35.Py, 52.30.+r, 52.55.Ez, 02.30.Hg

## I. INTRODUCTION

Recently, there has been growing interest in the behavior of rotating plasma systems, mainly due to the fact that tokamaks exhibit macroscopic flows after being heated by neutral beams.<sup>1</sup> Plasmas in other experimental devices were known for many years to rotate, for example in  $\theta$  pinches where the rotation is believed to be induced by end shorting of the electric field,<sup>2,3</sup> and also in field reversed plasmas, for reasons which are still controversial.<sup>4,5</sup>

Among the new questions that this phenomenon presents is the effect of rotation on the stability of the confined plasma. We will be concerned here with the linear stability of equilibrium states with flow, for plasmas described by the ideal magnetohydrodynamics model. The main difficulty in dealing with flow problems arises from the fact that they generally do not have a self-adjoint formulation like the static case,<sup>6</sup> and thus the powerful tool of an energy principle is not available. The most one can do is to use methods for Hamiltonian systems, as the plasma is known to be one.<sup>7</sup> Mainly due to this difficulty, flow problems were hardly treated until very recently and the few cases which were treated tended to be rather specialized, e. g., rotating  $\theta$  pinches<sup>8-10</sup> or multipoles with purely toroidal rotation.<sup>11</sup> A result of a more general character is the extension of Suydam's stability criterion<sup>12</sup> to the rotating plasma.<sup>13</sup>

In this work we will be concerned with the linear stability of general rotating equilibria. Because of the nonexistence of an energy principle we cannot hope to find conditions which are both necessary and sufficient for stability. However, it will be possible to derive sufficient conditions for stability as well as bounds on the spectrum in the complex plane. The bounds are necessary for an effective numerical investigation since they are the main tool for restricting the search for eigenmodes to a reasonably limited domain in the complex plane. Alternative devices, like properties of the nodal structure of the eigenfunctions, are not available in the non-self-adjoint case. One of the sufficient conditions we obtain is based on the conservation of circulation for a moving plasma. [All other conservation laws are automatically incorporated by a special choice of the perturbed density and magnetic field (5)]. Clearly, every conserved quantity restricts the growth of a perturbation and implies some special property of the equations, which we utilize. The improvement in stability, however, is effective only for symmetry preserving modes. Another sufficient condition, as well as

the bounds on the spectrum, are obtained through the use of circle theorems<sup>13,14</sup> which are a generalization of Howard's<sup>15</sup> estimate for incompressible fluids. This technique was recently utilized to obtain spectral bounds for a rotating  $\theta$  pinch<sup>10</sup> and for other configurations in fluid dynamics.<sup>16</sup>

Another objective of this work is to introduce the rotating screw-pinch as an important model for the investigation of the effects of flows. Due to the axial and azimuthal symmetries, the problem reduces to the study of an ordinary differential operator. It can actually be reduced to a single second order equation, even for magnetic fields and flows having both axial and poloidal components. This property was observed<sup>17</sup> in the static case, and helped make the screw-pinch the most fruitful case for studies of spectral properties of such equilibria.<sup>18</sup> We anticipate that the reduction of this problem to a manageable form will facilitate exhaustive analytical and numerical studies comparable to those carried out in the static case.<sup>19,20</sup>

In the next section we describe the equations of motion and reduce the rotating screw-pinch problem to a single second order equation. In Sec. III we derive a sufficient condition for stability based on the constants of the motion. The condition is then applied to the screw-pinch. Section IV describes an improved form of the circle theorem from which spectral bounds and another sufficient condition are obtained. Section V is devoted to a rather thorough application of the spectral bounds to the screw-pinch, commensurate with the importance we attach to this model. We show how these bounds can be computed by a minor modification of any numerical code which will be used in the investigation.

## II. THE EQUATIONS OF MOTION

The plasma is described using the ideal magnetohydrodynamics equations

$$\begin{aligned}\rho \mathbf{u}_t + \rho \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p &= \mathbf{J} \times \mathbf{B}, \\ \rho_t + \operatorname{div}(\rho \mathbf{u}) &= 0, \\ \mathbf{B}_t + \operatorname{curl}(\mathbf{B} \times \mathbf{u}) &= 0, \\ s_t + \mathbf{u} \cdot \nabla s &= 0, \\ \operatorname{div} \mathbf{B} &= 0, \quad \mathbf{J} = \operatorname{curl} \mathbf{B}, \quad p = p(\rho, s),\end{aligned}\tag{1}$$

where  $\mathbf{B}$ ,  $\mathbf{u}$ ,  $p$ ,  $\rho$ ,  $s$  are the magnetic field and the plasma velocity, pressure, density, and entropy, respectively. The

last equation is a prescribed equation of state. We also define

$$a^2 = \partial p(\rho, s) / \partial \rho, \quad \mathbf{B}_*^2 = \mathbf{B}^2 + \rho a^2, \quad (2)$$

$$\beta = \rho a^2 / B_*^2, \quad p_* = p + \mathbf{B}^2 / 2,$$

$a$  is the speed of sound. Given an equilibrium state with flow we linearize Eq. (1) about it. Frieman and Rotenberg<sup>7</sup> used the Lagrangian displacement vector  $\xi$  as the dependent variable to obtain the following linearized equation, which we describe in detail for later use,

$$\rho \xi_{tt} + 2A \xi_t - F \xi = 0, \quad (3)$$

where the operators  $A$  and  $F$  are

$$\begin{aligned} A \xi &= \rho \mathbf{u} \cdot \nabla \xi, \\ F \xi &= \nabla(\rho a^2 \nabla \cdot \xi + \xi \cdot \nabla p - \mathbf{B} \cdot \mathbf{Q}) + \mathbf{B} \cdot \nabla \mathbf{Q} + \mathbf{Q} \cdot \nabla \mathbf{B} \\ &\quad + \nabla \cdot (\rho \xi \mathbf{u} \cdot \nabla \mathbf{u} - \rho \mathbf{u} \mathbf{u} \cdot \nabla \xi) \end{aligned} \quad (4)$$

and  $\mathbf{Q} = \text{curl}(\xi \times \mathbf{B})$ . All coefficients are equilibrium quantities. Equation (3) is obtained by using the following relations between the perturbed Eulerian quantities (denoted by a subscript 1) and  $\xi$ :

$$\begin{aligned} \mathbf{B}_1 &= \text{curl}(\xi \times \mathbf{B}), \\ \rho_1 &= -\text{div}(\rho \xi), \\ p_1 &= -\xi \cdot \nabla p - \rho a^2 \text{div} \xi, \\ \mathbf{u}_1 &= \mathbf{u} \cdot \nabla \xi - \xi \cdot \nabla \mathbf{u} + \xi_t. \end{aligned} \quad (5)$$

$\xi(\mathbf{r}, t)$  expresses the displacement from the equilibrium trajectory of the fluid particle which would have been at the position  $\mathbf{r}$  at time  $t$  in the unperturbed motion. It was observed<sup>7</sup> that  $A$  is anti-Hermitian and  $F$  is Hermitian with respect to the inner product

$$(\xi, \eta) = \int \xi \cdot \eta^* d^3 \mathbf{r} \quad (6)$$

for twice differentiable vector functions with a vanishing normal component at the boundary of the plasma (which is assumed to be a flux surface). Equation (3) has the form of a general linear Hamiltonian system, where the momentum coordinates were eliminated in favor of  $\xi$ . The evolution of solutions in time can be found by considering a dependence  $\xi \sim \exp(-i\omega t)$ . We obtain the nonlinear eigenvalue problem in  $\omega$ ,

$$L_\omega \xi \equiv (\omega^2 \rho + 2\omega i A + F) \xi = 0. \quad (7)$$

Notice that by taking the inner product with  $\xi$  one obtains a quadratic equation for  $\omega$ , and its solution is not complex if

$$(F \xi, \xi)(\rho \xi, \xi) - (i A \xi, \xi)^2 \leq 0. \quad (8)$$

If this inequality holds for all  $\xi$  it can be concluded that the plasma is exponentially stable.<sup>7</sup>

A configuration of great simplicity is the rotating screw-pinch. The plasma is contained in a circular cylinder of length  $L$  (finite or infinite) and radius  $l$ . The two ends of the cylinder are identified to resemble a torus. All equilibrium profiles depend on the radial variable  $r$  only (in  $r, \theta, z$  coordinates) and the magnetic field and velocity have vanishing radial components. Thus  $\mathbf{B} = B_\theta \hat{\theta} + B_z \hat{z}$ ,  $\mathbf{u} = v \hat{\theta} + w \hat{z}$  and we denote  $v = r\Omega(r)$ . Such an equilibrium state has to satisfy the single equation

$$(p + \mathbf{B}^2/2)' + B_\theta^2/r = \rho v^2/r, \quad (9)$$

where the prime denotes  $d/dr$ . Notice that the axial velocity  $w$  does not enter into Eq. (9) so that an arbitrary axial flow could be superimposed on any equilibrium state to yield another such state. This of course is not true in a toroidal geometry where a centrifugal force associated with  $w$  will be present.

In the linearized system, the coordinates  $\theta, z$  are ignorable and we can assume a dependence  $\xi \sim \exp(i m \theta + k z - \omega t)$ . It was observed before<sup>21</sup> that in the linearized system (1) only two variables are differentiated in a direction normal to the equilibrium flux surfaces (the surfaces  $r = \text{const.}$  in the screw-pinch). Thus in our configuration only two equations contain derivatives while six others are algebraic. The algebraic equations can be used to eliminate six Eulerian variables and we obtain a  $2 \times 2$  ordinary differential system

$$A C \begin{pmatrix} \xi \\ \phi \end{pmatrix}' = A \begin{pmatrix} \xi \\ \phi \end{pmatrix}, \quad (10)$$

where the elements of the matrix  $A$  are

$$\begin{aligned} \lambda_{11} &= C [(2m/r^2)(B_\theta H + \rho \sigma v) - (1/r)A] \\ &\quad + (2/r) \rho^2 \sigma^2 B_\theta (\sigma H v + B_\theta \sigma^2) + (1/r) \rho^2 \sigma^2 v^2 A, \\ \lambda_{12} &= -[\rho^2 \sigma^4 + C(k^2 + m^2/r^2)], \\ \lambda_{21} &= -C [A^2 + A r ((1/r) p_*') - (4/r^2)(B_\theta H + \rho \sigma v)^2] \\ &\quad + [(1/r) \rho v^2 A + (2/r) \rho B_\theta (\sigma H v + B_\theta \sigma^2)]^2, \\ \lambda_{22} &= -\lambda_{11} - (1/r) A C, \end{aligned}$$

and where

$$\begin{aligned} H &\equiv \mathbf{k} \cdot \mathbf{B} \equiv m B_\theta / r + k B_z, \\ \sigma &\equiv \omega - \mathbf{k} \cdot \mathbf{u} \equiv \omega - (m \Omega + k w), \\ A &= H^2 - \rho \sigma^2, \quad C = B_*^2 (\beta H^2 - \rho \sigma^2), \end{aligned} \quad (11)$$

$\phi$  is the perturbed total pressure  $p_1 + \mathbf{B} \cdot \mathbf{B}_1$  and  $\xi$  is  $\xi_r$ , the radial component of  $\xi$ , and is related to the perturbed velocity by  $u_r = -i \sigma \xi$ .

For given  $k$  and  $m$ , Eq. (10) is an "eigenvalue" equation for  $\omega$ , where the boundary conditions to be satisfied are regularity of  $r \xi$  at  $r = 0$  and  $\xi = 0$  at the wall  $r = l$ . Notice that the coefficients in (10) are real except for  $\omega$ , and thus if  $\omega$  is an eigenvalue so is  $\omega^*$ . Also  $-\omega$ ,  $-\omega^*$  are eigenvalues for wavenumbers  $-k$  and  $-m$ . For stability it is sufficient to consider  $\omega$  in the upper half complex plane. Also observe the existence of a real continuous spectrum given by the range of  $\omega$  for which  $A = 0$  (the Alfvén continuum) or  $C = 0$  (the cusp continuum) anywhere in the plasma. For these values of  $\omega$  Eq. (10) becomes singular. This is analogous to the static case.<sup>18</sup>

System (10) can be further reduced to a single, second order equation in  $\xi$ . This can be achieved by multiplying the second equation in (10) by an integration function  $f$  such that it takes the form

$$(f A C \phi)' = f \lambda_{21} \xi, \quad (13)$$

and  $f$  should satisfy

$$(f A C)' = -f \lambda_{22}. \quad (14)$$

The first equation in (10) is now multiplied by  $f A C / \lambda_{12}$  and

differentiated. Eqs. (13) and (14) are used to eliminate  $\phi$  and  $f$ . The resulting equation has the form

$$\left( \frac{rAC}{\lambda_{12}} \xi' \right)' + r \left[ \frac{\det A}{AC\lambda_{12}} - \left( \frac{\lambda_{11}}{\lambda_{12}} \right)' \right] \xi = 0, \quad (15)$$

which has the associated quadratic form

$$Q(\xi) = \int_0^l (p|\xi'|^2 - q|\xi|^2) dr, \quad (16)$$

where  $p$  is the coefficient of  $\xi''$  and  $q$  the coefficient of  $\xi$  in (15). It is pointed out that the points where  $\lambda_{12} = 0$  are not singular points of Eq. (15), as is clear from Eq. (10).

We now make the observation, to be used later on, that Eq. (15) can be derived formally in a variational way. Let  $\omega$  be real for the time being, and consider the quadratic form associated with  $L_\omega$  in (7). The reality of  $\omega$  implies that  $L_\omega$  is Hermitian. Vary the quadratic form with respect to admissible  $\xi$  and find the Euler equation. We have  $\delta(L_\omega \xi, \xi) = (L_\omega \xi, \delta \xi) + (L_\omega \delta \xi, \xi) = 2\text{Re}(L_\omega \xi, \delta \xi) = 0$ . By taking  $i\delta \xi$  instead of  $\delta \xi$  we find  $\text{Im}(L_\omega \xi, \delta \xi) = 0$ . Thus  $L_\omega \xi = 0$ . In the screw-pinch case  $(L_\omega \xi, \xi)$  contains derivatives of  $\xi_r \equiv \xi$  only, but not of  $\xi_\theta, \xi_z$ . We find the stationary point of the quadratic form with respect to variations of  $\xi_\theta, \xi_z$  only, but not of  $\xi$ . This yields  $\xi_\theta$  and  $\xi_z$  as linear combinations of  $\xi$  and  $\xi'$ . For these values of  $\xi_\theta, \xi_z$  the variation of  $(L_\omega \xi, \xi)$  with respect to  $\xi$  will yield Eq. (15). Notice that although we used  $\omega$  real in the derivation, the same equation for  $\xi$  holds for complex  $\omega$  since the coefficients are known to be rational functions of  $\omega$ . We now claim that for  $\xi_\theta, \xi_z$  as found before,

$$(L_\omega \xi, \xi) = 2\pi Q(\xi). \quad (17)$$

This is clear since the two forms generate the same differential equation in  $\xi$  and thus can differ only by a constant multiple. The constant can be found by looking at the leading term in  $\omega^2$  for large  $\omega$ .

It will be useful for the implementation of the results described in Secs. IV and V, to write down the explicit form of  $(L_\omega \xi, \xi)$ . Letting  $(\xi, \eta, \zeta) = (\xi_r, i\xi_\theta, i\xi_z)$  we get a quadratic form with real coefficients (except  $\omega$ ).

$$(L_\omega \xi, \xi) = - \int (S_0 + S_1 + S_2) d^3r,$$

where for real  $\xi, \eta, \zeta$ ,

$$S_0 = B_*^2 \xi'^2 + \xi^2 [H^2 - \rho\sigma^2 - (B_*^2/r)' + r(B_\theta^2/r^2)'],$$

$$S_1 = (2/r)[(m/r)\eta + k\xi] \\ \times [\rho a^2(r\xi)' + \rho v^2\xi] + 2(\eta B_z - \xi B_\theta) \\ \times [(kB_\theta + (m/r)B_z)\xi/r \\ + ((m/r)B_z - kB_\theta)\xi'] + 4\rho\Omega\sigma\xi\eta,$$

$$S_2 = \rho a^2[(m/r)\eta + k\xi]^2 + (m^2/r^2 + k^2) \\ \times (\eta B_z - \xi B_\theta)^2 - \rho\sigma^2(\eta^2 + \xi^2).$$

### III. CIRCULATION INVARIANTS

In this section we assume that the equilibrium state consists of a family of closed flux surfaces, not necessarily nested, and labeled by a parameter  $\psi$ . We also require the entropy to be constant on each surface  $s = s(\psi)$ . (A form of the following results holds even if the entropy assumption is violated. See the remark at the end of the section.) Since entropy

and flux are carried by the fluid, the same properties hold for the time-dependent motion and indeed, even for the perturbed system, as can be derived from (5). It is known<sup>22</sup> that for such configurations  $\int_\psi \mathbf{u} \cdot \mathbf{B} d^3r$  is a constant of the motion, where the integral is understood to be taken over a volume enclosed by the same moving flux tube  $\psi$ . This conservation law is an analog of the conservation of circulation in fluid dynamics. We have, from the definition of  $\xi$ ,

$$\int_\psi \mathbf{u}(\mathbf{r} + \xi) \cdot \mathbf{B}(\mathbf{r} + \xi) d^3r \\ = \int_\psi \mathbf{u}_0(\mathbf{r}) \cdot \mathbf{B}_0(\mathbf{r}) d^3r + \text{initial perturbation}, \quad (18)$$

where here the subscript 0 denotes equilibrium quantities and the other quantities represent the nonlinear time dependent state. The initial perturbation term is time-independent and arises from the fact that perturbed states were not required to have the same  $\int_\psi \mathbf{u} \cdot \mathbf{B}$  as the equilibrium state. To first order in  $\xi$ ,

$$\mathbf{u}(\mathbf{r} + \xi) = \mathbf{u}(\mathbf{r}) + \xi \cdot \nabla \mathbf{u}(\mathbf{r}) = \mathbf{u}_0(\mathbf{r}) + \mathbf{u}_1(\mathbf{r}) + \xi \cdot \nabla \mathbf{u}_0(\mathbf{r})$$

and

$$d^3(\mathbf{r} + \xi) = (1 + \text{div} \xi) d^3r.$$

It follows from (5) after expanding  $\mathbf{B}(\mathbf{r} + \xi)$  and dropping the subscript 0,

$$\int_\psi [\mathbf{B} \cdot (\xi_t + \mathbf{u} \cdot \nabla \xi) + \mathbf{u} \cdot (\mathbf{B} \cdot \nabla \xi)] d^3r \\ = \text{initial perturbation}, \quad (19)$$

where all coefficients of  $\xi$  are equilibrium quantities. Using the equilibrium equations and also  $\mathbf{B} \cdot \nabla \psi = 0$  and  $\mathbf{u} \cdot \nabla \psi = 0$  (the equilibrium flow is within a flux surface), the last term under the integral can be integrated by parts and replaced by  $\mathbf{B} \cdot (\mathbf{u} \cdot \nabla \xi)$ . The initial perturbation term can be eliminated by differentiating Eq. (19) by  $t$ . We get

$$\int_\psi \mathbf{B} \cdot (\xi_{tt} + 2\mathbf{u} \cdot \nabla \xi_t) d^3r = 0. \quad (20)$$

A comparison with Eq. (3) shows

$$\int_\psi \mathbf{B} \cdot \mathbf{F}(\xi)/\rho d^3r = 0 \quad (21)$$

for every  $\psi$  and all admissible  $\xi$ . A direct verification of property (21) will be comforting. We use the symmetry of  $F$  to write

$$\int_\psi \mathbf{B} \cdot \mathbf{F}(\xi)/\rho d^3r \\ = \int_\psi \xi \cdot \mathbf{F}(\mathbf{B}/\rho) d^3r + \text{boundary terms}.$$

The boundary terms may be present since the normal component of  $\xi$  does not necessarily vanish on  $\psi$ . It can be easily seen directly, though, that  $F(\mathbf{B}/\rho) = 0$  and the boundary terms, which can be simply found from (4), also vanish. In fact, the same thing happens if  $\mathbf{B}$  is replaced by any vector field  $\mathbf{b}$  satisfying

$$\text{div} \mathbf{b} = 0, \quad \mathbf{b} \cdot \nabla \psi = 0, \quad (22)$$

$$\text{curl}(\mathbf{b} \times \mathbf{u}) = 0, \quad \text{curl}(\mathbf{b} \times \mathbf{B}/\rho) = 0, \quad (23)$$

and we have, corresponding to Eq. (20),

$$\oint_V \mathbf{b} \cdot (\xi_{,i} + 2\mathbf{u} \cdot \nabla \xi_i) d^3r = 0. \quad (24)$$

This property is related to the existence<sup>23</sup> of additional circulation invariants in magnetohydrodynamics. Equation (22) means that  $\mathbf{b}$  is a magnetic field-like vector field, having common flux surfaces with  $\mathbf{B}$ . To solve Eqs. (22) and (23), we first notice that  $\mathbf{b} = \mathbf{B}$  and  $\mathbf{b} = \rho\mathbf{u}$  are solutions. If  $\mathbf{u}$  is not parallel to  $\mathbf{B}$  we can write

$$\mathbf{b} = f\mathbf{B} + g\rho\mathbf{u}, \quad (25)$$

where  $f$  and  $g$  are scalar functions. However, from (23) we get  $(\mathbf{B} \times \mathbf{u}) \times \nabla f = 0$ , hence  $f = f(\psi)$ . Likewise  $g = g(\psi)$  and then  $\mathbf{b}$  is also divergence free. If  $\mathbf{u}$  is parallel to  $\mathbf{B}$ , the two equations in (23) have the same content. To find a general  $\mathbf{b}$  we write  $\mathbf{b} = f\mathbf{B} + g\mathbf{B} \times \nabla\psi$ . Equation (23) implies that  $g\mathbf{B}^2/\rho = h(\psi)$ , some function of  $\psi$ . Since  $\mathbf{b}$  is divergence free, we have  $\mathbf{B} \cdot \nabla f = -h(\psi)\text{div}(\rho\mathbf{B} \times \nabla\psi/\mathbf{B}^2)$ , and  $f$  is determined up to a function of  $\psi$  (for ergodic field lines), i. e.,

$$\mathbf{b} = k(\psi)\mathbf{B} + h(\psi)\mathbf{b}_1, \quad (26)$$

where  $\mathbf{b}_1$  is a particular solution for  $\mathbf{b}$  which has a  $\mathbf{B} \times \nabla\psi$  component and therefore is never parallel to  $\mathbf{B}$ . [Notice that  $\rho\mathbf{u}$  also must have the form (26) and therefore if it is parallel to  $\mathbf{B}$  at a point, it will be parallel over the entire flux surface.] We remark that if the equilibrium state is axisymmetric, say independent of a toroidal angle  $\theta$ ,  $\mathbf{b} = r\rho\hat{\theta}$  is a solution, where  $r$  is the cylindrical radial coordinate. This solution is suggested by conservation of angular momentum. Also, for static equilibria  $\mathbf{b} = \mathbf{J}$  is a solution. Our first conclusion is

**Theorem 1:** The point  $\omega = 0$  is an eigenvalue of Eq. (7) with infinite multiplicity.

*Proof:* It was already pointed out that  $F(\mathbf{b}/\rho) = 0$ . The theorem was known for the static case but our treatment reveals its origin, namely conservation of circulation. The existence of an  $\omega = 0$  eigenvalue does not therefore indicate bifurcation or marginal stability, at least within the present ideal plasma model.

It is highly suggestive that we try to eliminate these zeroes of  $F$  from consideration, that is, restrict ourselves to states which automatically satisfy the conservation of circulation. In order to do this it is convenient to introduce a new inner product

$$\langle \xi, \eta \rangle = \int \rho \xi \cdot \eta^* d^3r. \quad (27)$$

With respect to (27) we have the Hermitian operators  $F'$ ,  $A'$ , where

$$A' = (i/\rho)A, \quad F' = (1/\rho)F, \quad (28)$$

and the eigenvalue equation (7) can be rewritten as

$$(F' + 2\omega A' + \omega^2)\xi = 0. \quad (29)$$

Define now the subspace  $S$  which consist of all vectors  $\mathbf{b}/\rho$ , with  $\mathbf{b}$  as in (26). Let  $P$  denote the orthogonal projection operator associated with  $S$ . Namely,  $P\xi = f(\psi)\mathbf{B}/\rho + g(\psi)\mathbf{u}$  and  $f(\psi)$ ,  $g(\psi)$  are determined by the requirement  $\langle P\xi, \mathbf{b}/\rho \rangle = \langle \xi, \mathbf{b}/\rho \rangle$  for all  $\mathbf{b}$ .  $f$  and  $g$  can be easily found if we determine them on a surface by surface basis, and observe that  $d^3r = d\psi dS/|\nabla\psi|$ . We want

$$\oint P\xi \cdot \mathbf{b} \frac{dS}{|\nabla\psi|} = \oint \xi \cdot \mathbf{b} \frac{dS}{|\nabla\psi|}, \quad (30)$$

where  $dS$  is an area element on a flux surface. There are two independent  $\mathbf{b}$  on each surface, and (30) completely determines  $P\xi$ .

Clearly,

$$P^2 = P = P^*, \quad (31)$$

which is a restatement of the fact the  $P$  is a projection onto  $S$  in the direction orthogonal to  $S$ . The property  $F(\mathbf{b}/\rho) = 0$  implies  $F'P = 0$ . Taking the adjoint we have  $PF' = 0$ . To summarize,

$$PF' = F'P = 0. \quad (32)$$

Define the projection on the orthogonal complement of  $S$ ,  $Q = I - P$ , where  $I$  is the identity operator. We have  $Q = Q^2 = Q^*$  and  $QP = PQ = 0$ . Now multiply Eq. (29) by  $P$  and write  $\xi = P\xi + Q\xi$ . Because of (32) we have

$$\omega(2PA'P + 2PA'Q + \omega P)\xi = 0. \quad (33)$$

For stability purposes we are not interested in  $\omega = 0$  and it can be divided out. We now claim

$$PA'P = 0. \quad (34)$$

To prove it we recall that  $A' = i\mathbf{u} \cdot \nabla$  so that for any  $\xi$ ,  $A'P\xi = \mathbf{u} \cdot \nabla(\mathbf{b}/\rho)$  for some  $\mathbf{b}$ . Using (23) it can be directly shown that  $\int \mathbf{b}' \cdot \mathbf{u} \cdot \nabla(\mathbf{b}/\rho) d^3r = 0$  for any  $\mathbf{b}$ ,  $\mathbf{b}'$  as in (26). This shows that  $A'P\xi$  is orthogonal to  $S$  and thus proves (34). From (33) we now have for an eigenvector with  $\omega \neq 0$ ,

$$\omega P\xi = -2PA'Q\xi. \quad (35)$$

Now, multiplying (29) by  $Q$  we get [since  $QF' = (Q + P)F' = F'$ ],

$$F'\xi + 2\omega QA'(Q + P)\xi + \omega^2 Q\xi = 0,$$

or, using (35),

$$[(F' - 4QA'PA'Q) + 2\omega QA'Q + \omega^2]Q\xi = 0. \quad (36)$$

This is the restriction of Eq. (29) to the subspace orthogonal to  $S$ . Notice that it is still quadratic in  $\omega$ . Also, because of (34), and after denoting  $\eta = Q\xi$ , we have

$$[(F' - 4A'PA') + 2\omega QA'Q + \omega^2]\eta = 0, \quad (36')$$

and  $\eta$  automatically satisfies  $\eta = Q\eta$  if  $\omega \neq 0$ . The eigenvector  $\xi$  of Eq. (29) is obtained using (35), and is  $\xi = \eta - 2PA'\eta/\omega$ . An immediate result of (36') is, as in Eq. (8),

**Theorem 2:** A sufficient condition for stability is  $F + 4AP(1/\rho)A \leq 0$ . [We returned to the original notations of the operators and use the inner product (6) with respect to which  $P(1/\rho)$  is a symmetric operator.]

We now test our condition on the screw-pinch where we consider the mode which preserves both the axial and azimuthal symmetries, namely the  $m = 0$ ,  $k = 0$  mode.

**Theorem 3:** For the mode  $m = 0$ ,  $k = 0$  in the screw-pinch, the sufficient condition in Theorem 2 is also necessary for stability.

*Proof:* In this case, the effect of  $P$  is to eliminate the  $r$  component of  $\xi$ , while  $Q\xi = \xi_r(r)\hat{r}$ .  $\mathbf{u} \cdot \nabla Q\xi$  is then a vector with a  $\theta$  component only, so  $QA'Q = 0$  and Eq. (36') reduces

to an ordinary eigenvalue problem for a self-adjoint operator. The nonpositivity of this operator is necessary and sufficient for stability.

**Remark:** Even if the assumption of constant entropy on flux surfaces is violated, which is possible for nonergodic fluid trajectories, it is still true that  $F(\mathbf{b}/\rho) = 0$ , with  $\mathbf{b} = f(\psi)\rho\mathbf{u}$ . Thus Theorems 1 and 2 still hold, with  $P$  being the projection into this smaller subspace.

Our sufficient condition is somewhat reminiscent of the Frieman-Rosenbluth condition (8). The two criteria actually complement each other. To see this, observe that  $A$  and  $F$  are real operators unless there is symmetry and one considers a particular Fourier mode denoted, say, by a wavenumber  $m$ . For  $m = 0$ ,  $A$  and  $F$  are still real and considering a real test function  $\xi$ ,  $(A\xi, \xi) = 0$  with no imaginary part. For such vectors condition (8) reduces to  $(F\xi, \xi) \leq 0$ . Since  $F$  is also real the maximum of this quadratic form is attained for  $\xi$  real, so that condition (8) actually reduces to  $(F\xi, \xi) \leq 0$  for all  $\xi$ , or  $F \leq 0$ . For  $m \neq 0$ , however, the  $A$  term in the condition offers an improvement. Considering our new condition, we notice that for  $m \neq 0$  the right-hand side of Eq. (30) always vanishes [ $\oint \exp(im\theta) d\theta = 0$  for  $m \neq 0$ ] so  $P = 0$  and the condition reduces to  $F \leq 0$ . Improvement then is possible only for  $m = 0$ . Notice that in the screw-pinch case, where there are two symmetries, only the mode  $m = 0$ ,  $k = 0$  could be considered. Theorem 3 shows that occasionally (but most likely not always) the improvement achieved is maximal. We conclude that our condition should be used for symmetry preserving modes while condition (8) should be used for all other modes.

#### IV. CIRCLE THEOREMS

In this section we concern ourselves with the location in the complex plane of eigenvalues of the Hamiltonian equation (7).  $\rho$ ,  $iA$ , and  $F$  are assumed to be Hermitian operators in an inner product space with  $\rho$  invertible (not necessarily positive definite). Taking the inner product of (7) with  $\xi$  and solving for  $\omega$ , we have

$$\omega = \{ -(iA\xi, \xi) + [(iA\xi, \xi)^2 - (\rho\xi, \xi)(F\xi, \xi)]^{1/2} \} / (\rho\xi, \xi). \quad (37)$$

For a complex  $\omega$  one finds<sup>24</sup>

$$\operatorname{Re} \omega = -(iA\xi, \xi) / (\rho\xi, \xi), \quad (38)$$

$$|\omega|^2 = (F\xi, \xi) / (\rho\xi, \xi), \quad (39)$$

and the following estimates hold:

$$\inf_{\xi} -(iA\xi, \xi) / (\rho\xi, \xi) \leq \operatorname{Re} \omega \leq \sup_{\xi} -(iA\xi, \xi) / (\rho\xi, \xi), \quad (40)$$

$$\inf_{\xi} (F\xi, \xi) / (\rho\xi, \xi) \leq |\omega|^2 \leq \sup_{\xi} (F\xi, \xi) / (\rho\xi, \xi). \quad (41)$$

All complex eigenvalues  $\omega$  are contained in the strip (40) and in a concentric annulus centered at  $\omega = 0$ . Notice, however, that the lower bound in (41) may be negative or the upper bound infinite in which case some of the bounds are not useful. This will certainly be the case if  $\rho$  is not definite. Improvement of the previous results is possible by observing that the bilinear transformation

$$\omega = (az + b)/(cz + d), \quad ad - bc \neq 0 \quad (42)$$

yields another quadratic equation in  $z$  which, moreover, preserves the symmetry properties of the operator coefficients if  $a, b, c, d$  are real numbers. This is assumed from now on. The coefficients transform according to the linear transformation rules of quadratic forms, that is,

$$\begin{pmatrix} \rho & iA \\ iA' & F' \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} \rho & iA \\ iA & F \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad (43)$$

where here primes indicate the new operators in (7) after  $\omega$  is replaced by  $z$ . Estimates (40) and (41) can now be carried out for  $z$ . Since the transformation (42) maps and circles and lines into circles and lines, one gets (nonconcentric) annuli in the intersection of which all complex eigenvalues  $\omega$  must be contained. Notice that  $z = -\omega$  is a transformation which does not yield new bounds. Also it is possible to scale the coefficients in (42) without affecting anything. Hence we may restrict ourselves to real coefficients satisfying

$$ad - bc = 1. \quad (44)$$

To cover all possibilities, hopefully with some  $\rho'$  definite, involves a 3-parameter group of transformations. We remark that the transformation rule (43) applies also to the quadratic forms  $(F\xi, \xi)$  and the like, and that the Frieman-Rosenbluth condition (8) involves the determinant of the matrix in (43). The determinant is unchanged by the transformation considered so that the sufficient condition (8) is not improved.

The case of  $\rho$  positive definite is of particular interest. It is useful in this case to consider the special class of transformations

$$\omega = z + \alpha \quad (45)$$

with  $\alpha$  real. Equation (7) takes a form quadratic in  $z$ ,

$$[z^2\rho + 2z(iA + \alpha\rho) + L_{\alpha}]\xi = 0, \quad (46)$$

where  $L_{\alpha}$  is defined in (7). The bounds on  $\operatorname{Re} z$  yield (40) again, while an upper bound on  $|z|$  is

$$|z|^2 \leq \sup_{\xi} (L_{\alpha}\xi, \xi) / (\rho\xi, \xi) \equiv R^2(\alpha), \quad (47)$$

We neglect here the lower bound which is sometimes useful.<sup>10</sup> One then obtains the following results<sup>13,14</sup>:

**Theorem 4:** All complex eigenvalues  $\omega$  are contained in the intersection of all disks of center  $\alpha$  and radius  $R(\alpha)$  as defined in (47), for all real  $\alpha$ .

**Theorem 5:** A sufficient condition for exponential stability is

$$\inf_{\omega \text{ real}} \sup_{\xi} \frac{(L_{\omega}\xi, \xi)}{(\rho\xi, \xi)} \leq 0. \quad (48)$$

(We change our notation here from  $\alpha$  to  $\omega$ ).

Condition (48) is not necessary for stability in general. To gain a better understanding of this condition we first point out that if  $L_{\omega}$  is self-adjoint and  $\rho$  is positive definite, then  $R^2(\omega)$  is the upper edge of the  $\lambda$  spectrum of  $L_{\omega}$  defined by

$$L_{\omega}\xi = \lambda\rho\xi. \quad (49)$$

Consider now Eq. (7) in  $n$ -dimensional space with  $\rho$ ,  $iA$ ,  $F$  constant Hermitian matrices and  $\rho$  positive definite. Clearly

there are  $2n$  eigenvalues  $\omega$ . It is possible to find all the real ones by considering problem (49) for all real  $\omega$ .  $\omega$  is an eigenvalue if and only if  $\lambda_j(\omega) = 0$  for some  $j$ ,  $1 \leq j \leq n$ . If the largest eigenvalue  $\lambda_1$  becomes negative for some real  $\omega$ , then all curves  $\lambda_j(\omega)$  do so. Since for large  $|\omega|$ ,  $\lambda_j(\omega)$  behave like  $\omega^2$  they must have vanished at least twice, and therefore, there are exactly  $2n$  real eigenvalues  $\omega$  and stability follows (see Fig. 1(a)). The unnecessary of condition (48) is described in Figs. 1(b) and 1(c). Two eigenvalue curves  $\lambda_j(\omega)$  may intersect and exchange their labels, or one of the lower eigenvalues  $\lambda_j$ ,  $j > 1$ , may intersect the  $\omega$  axis more than twice even though  $\lambda_1$  does not intersect it at all. For the screw-pinch we will show, however, that the situation in Fig. 1(b) cannot occur.

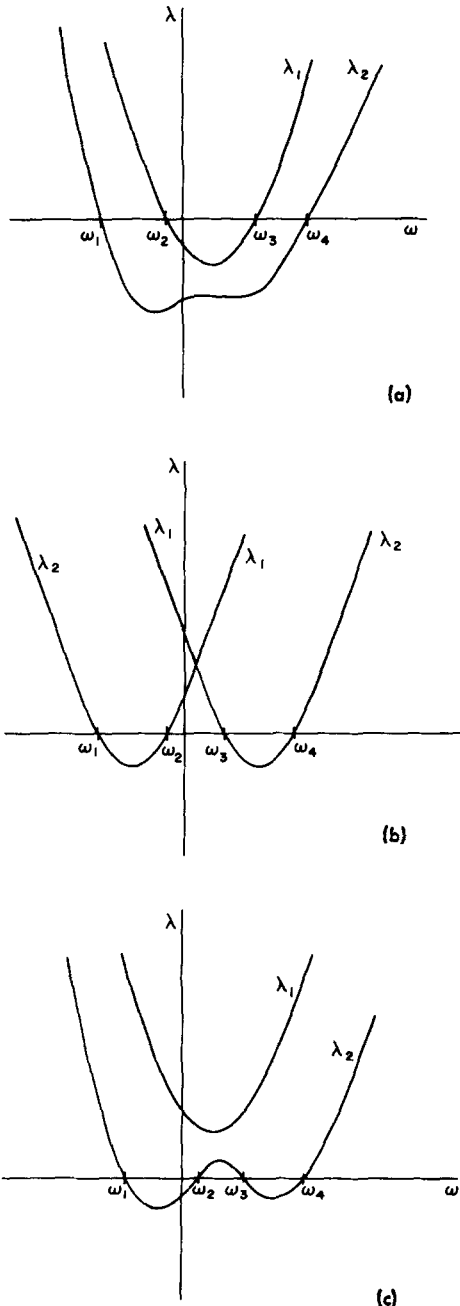


FIG. 1. Behavior of eigenvalues  $\lambda_j(\omega)$  of Eq. (49) with  $2 \times 2$  matrices in stable cases. (a) Sufficient condition (48) is satisfied. (b) and (c) The condition is violated.

## V. SPECTRAL BOUNDS FOR THE SCREW-PINCH

The rotating screw-pinch with sufficiently smooth plasma profiles presents a workable model for the application of the theory described in the previous section, as well as a possibility to check the closeness of the spectral bounds to reality. This section is devoted to the derivation of results which will facilitate such an application. The  $\lambda$ -eigenvalue problem (49) is the main tool of our discussion. A computation of the  $\lambda$ 's can be made possible in the screw-pinch case by observing that a change  $\omega^2 \rightarrow \omega^2 - \lambda$  in  $L_\omega$  has a form similar to (7) and will likewise reduce to a system like (10). Notice that only  $\omega^2$  is changed but not  $\omega$ . We can distinguish between the two by noting that  $\omega$  in  $L_\omega$  is multiplied by  $u$  components while  $\omega^2$  is not. The system (10) then should be modified such that  $\sigma^2 \rightarrow \sigma^2 - \lambda$  everywhere except for the last term in  $\lambda_{11}$  (and  $\lambda_{22}$ ), while  $\sigma$  is not changed. In particular  $A \rightarrow \hat{A} = H^2 - \rho(\sigma^2 - \lambda)$ ,  $\rho^2 \sigma^4 \rightarrow \rho^2(\sigma^2 - \lambda)^2$ . We will denote modified expressions by adding a circumflex to the original notation, e. g., Eq. (10).  $\omega$  will be considered a real parameter. For convenience we cast some of the following results as claims.

**Claim 1:** For fixed wave numbers  $m, k$  and for fixed real  $\omega$ ,  $L_\omega$  is essentially self-adjoint and bounded from above while the operator  $A$  is bounded.

**Proof:**  $L_\omega$  is a second order Hermitian ordinary differential operator and  $(L_\omega \xi, \xi)$  involves derivatives of  $\xi$  only but not of  $\xi_\theta, \xi_z$ . Since we imposed two boundary condition on  $\xi$ , self-adjointness can be shown in the same way that it is proved for more standard regular ordinary differential operators. (One shows that the Friedrichs self-adjoint extension of  $L_\omega$  requires the same boundary conditions as we already imposed.<sup>25</sup>) The highest order term in  $\xi'$  in  $(L_\omega \xi, \xi)$  is  $-2\pi \int r B_*^2 |\xi'|^2 dr \leq 0$ . Thus it can be maximized with respect to  $\xi'$ , considered as independent of  $\xi$ . What is left is an algebraic quadratic form in  $\xi$  which is bounded. The inefficient circle bounds based on this approximation appear in Ref. 13. The operator  $A$  reduces to a matrix multiplication and therefore is bounded. Indeed  $iA\xi = -\rho \mathbf{k} \cdot \mathbf{u} \xi + i\rho \Omega (-\xi_\theta \hat{r} + \xi \theta)$ . Bounds (40) are given in terms of the  $\lambda$  spectrum for  $-iA\xi = \lambda \rho \xi$ . For each point in the plasma,  $-iA$  has three  $\lambda$  eigenvalues, namely  $\mathbf{k} \cdot \mathbf{u}$  and  $\mathbf{k} \cdot \mathbf{u} \pm \Omega$ . The spectrum of  $-iA$  is the range of these values. It follows from (40) that

$$\text{Min}_r(\mathbf{k} \cdot \mathbf{u} - |\Omega|) \leq \text{Re} \omega \leq \text{Max}_r(\mathbf{k} \cdot \mathbf{u} + |\Omega|). \quad (50)$$

**Claim 2:**  $L_\omega$  has a  $\lambda$ -continuous spectrum consisting of all points  $\lambda = \sigma^2 - \beta H^2/\rho$ ,  $\lambda = \sigma^2 - H^2/\rho$  for any  $r$  in the plasma.

**Proof:** This is seen most directly from the analogy with the  $\omega$  spectrum.<sup>18</sup> The continuous spectrum is visible in (10) when letting  $\hat{A}$  and  $\hat{C}$  vanish. Note that since  $\beta < 1$ , the upper edge  $\lambda_c$  of the  $\lambda$ -continuous spectrum is given by

$$\lambda_c(\omega) = \text{Max}_r(\sigma^2 - \beta H^2/\rho). \quad (51)$$

**Claim 3:** Eigenvalue curves  $\lambda(\omega)$  outside the  $\lambda$ -continuous spectrum do not intersect each other.

**Proof:** The  $\lambda$  eigenvalues can be obtained from the sec-



second order ordinary differential scalar equation (15) when a solution satisfies the two boundary conditions. Intersection of two curves  $\lambda(\omega)$  at  $\omega = \omega_0$ ,  $\lambda = \lambda_0$  means that at this point two independent solutions of (15) vanish at the boundary. Hence every solution (their linear combination) should do so, which is impossible since one can prescribe an arbitrary value for a solution there.

**Claim 4:** The highest eigenvalue curve  $\lambda_1(\omega)$  is convex as long as  $\lambda_1(\omega) > \lambda_c(\omega)$ .

**Proof:**  $\lambda_1(\omega)$  is an analytic function of  $\omega$ . Differentiate the relation  $L_\omega \xi = \lambda_1 \rho \xi$  twice by  $\omega$  and denote such derivatives by dots.

$$L_\omega \dot{\xi} + 2(\rho\omega + iA)\dot{\xi} = \lambda_1 \rho \dot{\xi} + \dot{\lambda}_1 \rho \xi,$$

$$L_\omega \ddot{\xi} + 4(\rho\omega + iA)\dot{\xi} + 2\rho\ddot{\xi} = \lambda_1 \rho \ddot{\xi} + 2\dot{\lambda}_1 \rho \dot{\xi} + \ddot{\lambda}_1 \rho \xi.$$

Take the inner product of the second equation with  $\xi$  and of the first equation with  $2\dot{\xi}$  and subtract. It follows that

$$\dot{\lambda}_1(\rho\xi, \xi) = 2(\rho\xi, \dot{\xi}) + 2[\lambda_1(\rho\dot{\xi}, \xi) - (L_\omega \dot{\xi}, \xi)].$$

Since  $\lambda_1$  is the uppermost point in the spectrum of  $L_\omega$ , the bracket term is nonnegative and we have

$$d^2\lambda_1/d\omega^2 \geq 2 \quad \text{for } \lambda_1 > \lambda_c. \quad (52)$$

We have obtained the following result

**Theorem 6:** Let  $R^2(\omega) = \max\{\lambda_1(\omega), \lambda_c(\omega)\}$  for  $\omega$  real, where  $\lambda_1$  is the largest isolated eigenvalue of  $L_\omega$ . If  $R^2(\omega) \leq 0$  for any real  $\omega$ , the plasma is stable. If this condition is not satisfied then every complex  $\omega$  eigenvalue of Eq. (7) is contained in the intersection of all circles with center  $\omega$  and radius  $R(\omega)$  for all real  $\omega$ . If  $\lambda_1(\omega) > \lambda_c(\omega)$  for all real  $\omega$ , there is a *unique* circle of smallest radius. If  $\lambda_c(\omega) > \lambda_1(\omega)$  for all  $\omega$ , the circle (or circles) of smallest radius  $R$  is found by minimizing  $\lambda_c(\omega)$  and

$$R^2 = \min_{\omega} \max_r (\sigma^2 - \beta H^2/\rho).$$

The following discussion deals with the question of how to decide the location of  $\lambda_1$  relative to  $\lambda_c$ .

**Claim 5:** For fixed real  $\omega$  and in the region  $\lambda > \lambda_c(\omega)$ , solutions of Eq. (15) oscillate slower as  $\lambda$  increases.

**Proof:** By slower oscillation it is meant that all zeroes of a solution which vanish at a fixed point, move monotonically away from that point as  $\lambda$  increases. We prove the claim by noting that the quadratic form  $\hat{Q}(\xi)$  corresponding to (15) satisfies, similar to Eq. (17),  $2\pi\hat{Q}(\xi) = (L_\omega \xi, \xi) - \lambda(\rho\xi, \xi)$ , where  $\xi_\theta, \xi_z$  take their stationary value for given  $\xi$ . For  $\lambda > \lambda_c$  it can be shown directly that the stationary point is actually a maximum. Clearly  $L_\omega - \lambda\rho$  decreases monotonically with  $\lambda$ , and thus so does  $\hat{Q}(\xi)$ . The oscillation result now follows similar to the usual one for ordinary differential equations.

In order to find whether  $\lambda_1 < \lambda_c$  or not, it is sufficient to decide whether  $L_\omega - \lambda_c(\omega)\rho$  is negative semidefinite. If it is negative then  $\lambda_c$  is the upper edge of the spectrum. Notice that for  $\lambda = \lambda_c$ , the coefficient  $\hat{C}$  in (10) vanishes somewhere in the plasma and the equation becomes singular. In fact, since  $\lambda_c$  has the maximum property (51),  $\hat{C}$  will have in general a double zero. Yet, the singularity will still be a regular singularity as can be seen from Eq. (15) after we note that the matrix  $A$  can be written as  $A = A_0 + CA_1$  and  $\det A_0 \equiv 0$ , so

that  $\det A$  is divisible by  $C$ . For the case of a regular singularity at  $r_0$  we know that solutions behave like  $(r - r_0)^{c_i}$  ( $i = 1, 2$ ). If the  $c_i$  are complex, solutions oscillate rapidly near  $r_0$ . We can now state the following

**Theorem 7:** For a fixed real  $\omega$ ,  $\lambda_c$  as defined in (51) is the uppermost point in the spectrum of  $L_\omega$  if and only if corresponding to all increasingly ordered points  $r_1, \dots, r_s$  for which  $\hat{C}$  vanishes in the plasma (with  $\lambda = \lambda_c$ ), the indices  $c_i(r_j)$  ( $i = 1, 2, j = 1, \dots, s$ ) are real and no solution of Eq. (15) vanishes more than once in any of the intervals  $(0, r_1), (r_s, l), (r_j, r_{j+1})$  for  $j = 1, \dots, s-1$ .

**Proof:** The theorem is an analog of the well-known Newcomb theorem<sup>26</sup> for the stationary pinch. The nonoscillation requirement of the theorem implies  $\hat{Q}(\xi) \leq 0$ , hence  $L_\omega - \lambda_c \rho \leq 0$ . Notice that the theorem applies only if  $\hat{C}$  has at most a double zero at any  $r_j$ , but not higher. If  $\hat{C}$  has a simple zero (the maximum in (51) is attained at the boundary) the singularity is still regular and the theorem holds.

From a practical point of view it may be more advisable to consider  $\lambda$  slightly larger than  $\lambda_c$  and then use claim 5 to track down  $\lambda_1$  numerically. That is, if a solution of (15) which satisfied the regularity condition at  $r = 0$  (equivalent to having a zero there) vanishes before  $r = l$ , take  $\lambda$  larger until the first zero reaches  $r = l$  and vice versa. We therefore do not pursue further the question of the value of the indices  $c_i$  in the last theorem or the question of optimizing the upper bound. We do discuss, however, our sufficient condition (48) for a particular point  $\omega = \omega_0$  which is important in the derivation of Suydam's condition for rotating equilibria<sup>13</sup> and is also exceptional with respect to the previous discussion on  $\lambda_c$ . This is the point for which  $A$  and  $C$  both vanish at some point  $r = r_0$ . It follows that

$$H(r_0) \equiv \mathbf{k} \cdot \mathbf{B}(r_0) = 0, \quad \omega_0 = \mathbf{k} \cdot \mathbf{u}(r_0). \quad (53)$$

We assume that the wavenumbers are such that  $\mathbf{k} \cdot \mathbf{B}$  does indeed vanish somewhere in the plasma. We also assume that  $h$  and  $\alpha$  do not both vanish, where

$$h = H'(r_0), \quad \alpha = (\mathbf{k} \cdot \mathbf{u})'(r_0). \quad (54)$$

Equation (53) can be used to eliminate one of the wavenumbers from the definitions of  $h$  and  $\alpha$ . Let  $\mu$  be the pitch of the magnetic field,

$$\mu = B_\theta/rB_z. \quad (55)$$

We have

$$h = -kB_z\mu'/\mu, \quad \alpha = k(\omega' - \Omega'/\mu). \quad (56)$$

$\mu'$  is the magnetic shear while  $\alpha$  is seen to be essentially the shear of the flow in a direction perpendicular to  $\mathbf{B}$ .

We would like to consider the possibility of  $\lambda_c(\omega_0) = 0$ . Notice that  $\sigma^2 - \beta H^2/\rho$  has a stationary point at  $r_0$  for  $\omega = \omega_0$ . It will be a local maximum if  $\beta h^2 - \rho\alpha^2 > 0$ , which we assume. This can be written as

$$\beta B_z^2 \mu'^2 > \rho(\Omega' - \mu\omega')^2, \quad (57)$$

and our assumption refers to the point  $r = r_0$ . In order to have  $\lambda_c(\omega_0) = 0$ ,  $r_0$  must be the global maximum in (51). If  $H$  vanishes in more than one point, say at  $r = r_j$  ( $j = 1, \dots, s$ ), then corresponding to each point we have an  $\omega_j$  as in (53). If not all the  $\omega_j$  are equal we get  $\lambda_c(\omega_j) \neq 0$  for all  $j$ , since at some



point  $r_i$ ,  $\sigma \neq 0$  while  $H = 0$ . Equality of all the  $\omega_j$  can be guaranteed if the flow is parallel to  $\mathbf{B}$  up to a superimposed rigid rotation of the plasma column.

To apply the sufficient condition (48), consider Eq. (15) (i. e., Eq. (15) with  $\lambda = 0$ ). Denoting  $x = r - r_0$  we notice that near the singular point  $\lambda_{11} = O(x^3)$ ,  $\lambda_{12} = O(x^2)$ ,  $\lambda_{21} = O(x^4) = \lambda_{22}$ , and  $A = O(x^2) = C$ . Thus the coefficients of the equation have  $p = O(x^2)$ ,  $q = O(1)$  and the singularity is again regular. The indicial equation is of the form  $c(c+1) + I = 0$  and the solutions do not oscillate rapidly if  $I < \frac{1}{4}$ . This is Suydam's condition<sup>13</sup> for local stability when (57) is satisfied. Since the explicit form of the condition is not readily available, we bring it here. Define

$$\phi = (\sqrt{\rho})(\Omega' - \mu w')/(\mathbf{B}_z \mu'), \quad \psi = (\sqrt{\rho})v/B_\theta.$$

The condition requires

$$\begin{aligned} & \frac{1}{8} B_z^2 (1 - \phi^2) \left( \frac{\mu'}{\mu} \right)^2 - \phi \psi B_z^2 \frac{\mu'}{\mu} + \frac{1}{2} r^2 \frac{B_\theta^2}{B_z^2} (\rho \Omega'^2) + p' \quad (58) \\ & > \frac{2}{\beta - \phi^2} \frac{B_\theta^2}{r} \left[ (1 - \beta) \left[ \frac{1}{4} \psi^4 (1 - \phi^2) - \psi^3 \phi + \frac{3}{2} \phi^2 \psi^2 \right] \right. \\ & \quad \left. + \beta \left[ \frac{1}{2} \psi^2 (1 + \phi^2) - 2\phi\psi + \phi^2 \right] \right]. \end{aligned}$$

When we let the flow vanish, (58) reduces to the classical Suydam condition<sup>12</sup>

$$\frac{1}{8} B_z^2 \left( \frac{\mu'}{\mu} \right)^2 + p' > 0. \quad (59)$$

If the flow is purely axial ( $\psi = 0$ ) which is the expected case for toroidal devices,<sup>1</sup> (58) reduces to

$$\frac{1}{8} B_z^2 \left( \frac{\mu'}{\mu} \right)^2 (1 - \phi^2) + p' > \frac{2\beta\phi^2}{\beta - \phi^2} \frac{B_\theta^2}{r} \quad (60)$$

and is more difficult to satisfy than (59) since  $\beta > \phi^2$  by (57). We can state the following result:

**Theorem 8:** Let  $\mathbf{k} \cdot \mathbf{B}$  vanish in the plasma at  $r = r_j$  ( $j = 1, \dots, s$ ) and  $\omega_j = \mathbf{k} \cdot \mathbf{u}(r_j)$ . Assume that all the  $\omega_j$  ( $j = 1, \dots, s$ ) are equal (to  $\omega_0$ ) and that  $(\omega_0 - \mathbf{k} \cdot \mathbf{u})^2 - \beta H^2/\rho$  is everywhere negative except at  $r = r_j$ . Then it is necessary for stability that condition (58) hold at  $r = r_j$  ( $j = 1, \dots, s$ ) and it is sufficient for stability that no solution of Eq. (15) with  $\omega = \omega_0$  vanishes more than once in any of the subintervals  $(0, r_1)$ ,  $(r_i, l)$ ,  $(r_j, r_{j+1})$ ,  $j = 1, \dots, s - 1$ .

*Proof:* The necessary part was proven in Ref. 13 (without the maximality assumptions). The sufficiency part is a direct result of our sufficient condition (48) as described in the proof of Theorem 7. Notice that the violation of (58) implies that every solution vanishes infinitely many times in the appropriate subintervals. An interesting aspect of this theorem is that in the case of the static pinch, the sufficient condition is also *necessary* for stability.<sup>26</sup>

As a final topic of this section we discuss the dependence of our circle bounds on  $m$  and  $k$ . From the definition of  $\lambda_c$  we find

$$\lambda_c(s\omega; sm, sk) = s^2 \lambda_c(\omega; m, k) \quad (61)$$

for any number  $s$ , which is assumed from now on to be positive. Notice that the matrix  $\hat{A}$  in Eq. (10) is homogeneous of degree four as a function of  $m, k, \omega, \sqrt{\lambda}$ , except for the "exceptional" term  $-\hat{C}\hat{A}^2$  in  $\lambda_{21}$ , which is of degree six. As a result,  $\hat{p}$  of Eq. (15) is homogeneous of degree zero while

$\hat{q} = \hat{q}_0 + r\hat{A}$ , where  $\hat{q}_0$  is also of degree zero while  $\hat{A}$  is of course of degree two. Introducing  $s$  into the equation as a coefficient of  $m, k, \omega, \sqrt{\lambda}$  wherever these parameters appear and assuming  $\lambda \geq \lambda_c$  (hence  $\hat{A} > 0$ ), we find that  $\hat{Q}(\xi; s)$  in (16) is monotone decreasing in  $s$ . Denoting by  $R(m, k)$  the radius of the circle of smallest radius (possibly zero) after minimizing over  $\omega$ , we can now show

**Theorem 9:**  $R(sm, sk)/s$  is a nonincreasing function of  $s$ .

*Proof:* Assume that for  $m_0 = s_0 m$ ,  $k_0 = s_0 k$ , the smallest circle is centered at  $\omega = \omega_0$  with radius  $R_0$ ,  $\lambda_0 = R_0^2$  and  $\hat{Q}_0$  in (16) (with obvious notation) is nonpositive. For  $s > s_0$  and  $m' = sm$ ,  $k' = sk$ , consider  $\omega' = (s/s_0)\omega_0$  and  $\lambda' = \lambda_0(s/s_0)^2$ .  $\hat{Q}' \leq \hat{Q}_0$  and  $\lambda$  should be made smaller (unless  $\lambda = \lambda_c$ ) to restore the upper bound of  $\hat{Q}$  to zero. This proves the theorem.

**Theorem 10:** If a sufficient condition based on our circle estimates is satisfied for  $m = 1$  and all  $k$ , and for  $m = 0$ ,  $k \rightarrow 0$ , then the screw-pinch is stable to all modes.

*Proof:* This theorem is the counterpart of Theorem 1, Ref. 26 for the static pinch. The assertions of this theorem for  $m > 1$  follow from Theorem 9 by taking  $s = m$  and considering  $R(m, k) = R(s, sk/m)$ . If  $m = 0$ , it similarly follows that if  $R(0, k_1) = 0$ , it will also vanish for all  $|k| > |k_1|$ . Thus, the only mode to consider (except  $k = 0$  which is a special case) is either  $k_1 = 2\pi/L$  in a finite cylinder, or  $k \rightarrow 0$  in the infinite cylinder. The limit  $k \rightarrow 0$  should be taken by letting  $s = k$ , using  $s\omega$  and  $s^2\lambda$  instead of  $\omega$  and  $\lambda$ , and then taking  $s \rightarrow 0$ . This amounts to using Eq. (15) with  $k = 1, \omega$  and  $\lambda$  still appear and the exceptional term  $\hat{C}\hat{A}^2$  in  $q$  is dropped. If the circle estimates imply stability in this limit, it follows from Theorem 9 that all modes  $m = 0, k \neq 0$  are stable. However, we can also show that the mode  $m = 0, k = 0$  will be stable in this case. Indeed, stability of the  $m = 0, k \rightarrow 0$  mode is shown by finding some  $\omega$  and proper  $\lambda$  for which the limiting  $\hat{Q}$  in (16) is negative definite. Thus it will remain negative if we let  $\lambda \rightarrow +\infty$ . This last limit however is identical with what is obtained from (16) by setting  $k = 0, m = 0$ , and then  $\omega = 0$ . It is already known from Sec. III that the mode  $k = 0, m = 0$  has a necessary and sufficient condition for stability, which is precisely the nonpositivity of the quadratic form we obtained. This completes the proof.

To conclude this section we summarize its practical consequences. A numerical calculation of  $\omega$  eigenvalues and of our circle estimates will be facilitated by introducing two additional parameters to Eq. (10). The first is  $\lambda$ , as described at the beginning of the section, and the second is  $s^2$  as a coefficient of  $\hat{C}\hat{A}^2$  in  $\lambda_{21}$ .  $s$  will have the effect of enabling one to consider only modes with  $m = 0$  or  $m = 1$  while all other modes  $(m, k)$  can be replaced by  $m \rightarrow 1, k \rightarrow k/m, s = m$ . The benefit of this change is that the search for  $\omega$  is done over a region smaller by a factor  $1/s$  than in the actual problem. This is useful in particular if the actual complex eigenvalues  $\omega$  are sought (in which case  $\lambda$  is set to 0). Of course, once an  $\omega(s)$  eigenvalue is found in this way, it should be multiplied by  $s$  to correspond to its actual value. The computation should be first run for real  $\omega$  in the range (50) (shrunk by a factor  $s$ ). One should search for the first eigenvalue  $\lambda$  which will result in a circle bound on all complex eigenvalues. The  $\lambda$  parameter should then be set to 0 and the search for complex  $\omega$  eigenvalue should be carried out inside that circle.

## ACKNOWLEDGMENTS

The author wishes to thank Professor Harold Weitzner for introducing him to this area of plasma physics, and for many useful discussions. The author is also grateful to Dr. Abraham Kadish for his encouragement.

This work was supported by U. S. Department of Energy contract No. DE-AC02-76ER03077.

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